

Implicitly learning token semantics

Gary Nan Tie, Mar 3, 2026

Abstract

A fast implicit recurrent deep learning model of semantic attention is proposed for the purposes of large language model next-token and sequence-to-sequence learning.

Introduction

The implicitRNN model [1] functions similarly to a vanilla RNN, processing sequence inputs one by one. For each time step i , where $t_i \in \mathbb{R}^p$ represents the i -th element in a sequence $(t_1, t_2, \dots, t_m)$. The model input u_i for implicitRNN is a concatenation of t_i and the previous hidden state h_{i-1} , akin to a vanilla RNN. The implicit prediction rule for implicitRNN is as follows:

$$h_0 = 0; x_0 = 0;$$

$$u_i = [t_i \ h_{i-1}]$$

$$x = \Phi(Ax + Bu_i) \text{ (equilibrium equation)}$$

$$\hat{y}_i(u_i) = Cx + Du_i \text{ (prediction equation)}$$

$$h_i = \hat{y}_i(u_i).$$

In this setup, the recurrent layer is replaced with an implicit structure comprising the equilibrium and prediction equations, where we learn matrices A , B , C , D and fixed points x .

Let score be a desired similarity function, like a kernel for example. Input u_i corresponding to token t_i , generates a unique fixed point s_i via the equilibrium equation. The hidden states h , get updated via the prediction equation. Akin to transformer attention looking at query-key similarity, fix i , normalize $\text{score}(h_i, s_j)$ into probabilities, for hidden state h_i and fixed points s_j , to create a probability distribution over tokens, and output the most likely token. LLMs typically minimize perplexity. The wrinkle here is we can also regularize with the average Renyi α -entropy of these distributions. This enables robust next-token learning. Moreover, if we think of the latent tokens s_j as possible meanings of the hidden state h_i , then lowering average Renyi α -entropy sharpens and aligns semantics. Similar considerations (for $\text{score}(h_i, h'_j)$, state h_i corresponding to target token t_i and state h'_j corresponding to source token t'_j) apply to sequence-to-sequence translation.

Note that perplexity attends to syntactic fidelity, while average Renyi α -entropy attends to semantic fidelity. The objective of a token learning problem can be tailored to any desired emphasis between the two.

Implicitly learning token semantics

Input: $\begin{cases} \text{target tokens } t_1, t_2, \dots, t_m \\ \text{source tokens } t'_1, t'_2, \dots, t'_n \end{cases}$

State Machine

initial state $h_0 = 0$

next state $h_i = f_s(h_{i-1}, t_i)$

output $O_{i+1} = f_o(h_i)$, $i \in [m]$

Implicit RNN

$h_0 = 0$, $s_0 = 0$

$$u_i = \begin{pmatrix} t_i \\ h_{i-1} \end{pmatrix}$$

$$s_i = \varphi(A s_i + B u_i)$$

$$\hat{y}_i(u_i) = C s_i + D u_i$$

$$h_i = \hat{y}_i(u_i) = f_s(h_{i-1}, t_i)$$

Given input token t_i , define output

$$o_{i+1} = t_{j_i^\beta} = f_o(h_i)$$

where $\beta_{ij} = \text{softmax score}(h_i, s_j)$
 $j \in [m]$

and $j_i^\beta = \arg \max_{j \in [m]} \beta_{ij}$,

score a desired similarity function eg a kernel.

Define average Renyi α -entropy of $\{t_i\}$

$$E(\{t_i\}) = \frac{1}{m} \sum_{i=1}^m H_\alpha^i,$$

where $H_\alpha^i = H_\alpha(\{\beta_{ij}\}_{j=1}^m)$, H_α Renyi α -entropy.

Similarly for $\{t'_j\}_{j=1}^n$, define β'_{ij}

Next-token learning problem

Let $\lambda > 0$, learn matrices A, B, C, D

and fixed points s_i to minimize

$$\{ \text{Perplexity}(o_2, \dots, o_{m+1}) + \lambda E(\{t_i\}) \}$$

Next-token prediction

Initialization: let t_1 = the empty token
or a seed token

Given $[t_1]$, define $t_2 = o_2$

Given $[t_1, t_2]$, define $t_3 = o_3$

Given $[t_1, t_2, t_3]$, define $t_4 = o_4$

and so on, autoregressively generating the next token.

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Sequence-to-sequence learning

Define $\gamma_{ij} = \text{soft max}_j \text{score}(h_i, h'_j)$, $i \in [m]$, $j \in [n]$

and $j_i^* = \arg \max_{j \in [n]} \gamma_{ij}$

Given input t_i , $i \in [m]$

define output $o_i = t'_{j_i^*}$

So $t_1, \dots, t_m$ is translated to $o_1, \dots, o_m$

Let $E(\{t_i\}, \{t'_j\}) = \frac{1}{m} \sum_{i=1}^m H_\alpha^i$, where $H_\alpha^i = H_\alpha(\{\gamma_{ij}\}_{j=1}^n)$

seq-to-seq learning problem

Let $\lambda_1, \lambda_2, \lambda_3 > 0$,

learn A, B, C, D and s_i for $\{t_i\}$ and

A', B', C', D' and s'_j for $\{t'_j\}$ and

minimize $\{ \text{Perplexity}(o_1, \dots, o_m) +$

$$\lambda_1 E(\{t_i\}) + \lambda_2 E(\{t'_j\}) +$$

$$\lambda_3 E(\{t_i, \{t'_j\})$$

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References

- [1] 'The Extrapolation Power of Implicit Models'
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Max Emerling, Ashwin Ganesh, Laurent El Ghaoui
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